A Sharp Measure Obstruction for Blei's $Q_A^{(t)}$ Riesz Products

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Status

This packet records a candidate counterexample, likely valid. It answers the measure-existence part of Question 5.14 in Ron Blei's *From Parseval to Grothendieck and a little beyond, part I* [1]. The result is slightly stronger than a single counterexample: for $2 < t \le \infty$, $Q_A^{(t)}(y)$ represents a finite measure exactly for the square-summable vectors $y \in \ell^2(A)$.

Source Passage

Blei's Remark 5.13 explains that, outside the case $s = t = 2$, the third Parseval-type expression involving $Q_A^{(t)}$ is defined by the sums on the right, rather than by a bona fide integral. The following question is then posed on PDF page 49.

(5.98)

where all sums on the right are absolutely convergent.

Remark 5.13 (bona fide integrals?). *The left sides of the first two lines of* (5.98) *are well-defined Lebesgue integrals. In the case $s = t = 2$, the left side of the third line is a bona fide integral with respect to*

$$Q_A^{(2)}(\mathbf{y})(d\omega) = Q_A^{(2)}(\mathbf{y})(\omega)\ \mathbb{P}_A(d\omega),$$

but otherwise, for $1 \leqslant s < 2 < t \leqslant \infty$, the left side is defined by the sums on the right.

Question 5.14. *What can be said about $Q_A^{(t)}(\mathbf{y})$ for $\mathbf{y} \in l^t_{\mathbb{R}}(A)$ and $t \in (2, \infty]$? E.g., does $Q_A^{(t)}(\mathbf{y})$ represent a measure?*

5.4. **More interpolants.** For $A' \supset A$, let

In the source notation, the question asks: what can be said about $Q_A^{(t)}(y)$ for $y \in \ell^t_{\mathbb{R}}(A)$ and $t \in (2, \infty]$? In particular, does $Q_A^{(t)}(y)$ represent a measure?

Definitions

Let A be an index set and let

$$\Omega_A = \{-1, 1\}^A$$

with product Haar probability measure $\mathbb{P}_A$. For $\alpha \in A$, write $r_\alpha(\omega) = \omega(\alpha)$, and for a finite set $F \subset A$, write

$$r_F = \prod_{\alpha \in F} r_\alpha.$$

These r_F's are the Walsh characters of Ω_A.

For $2 < t \le \infty$ and a non-zero real vector $y \in \ell^t(A)$, put

$$R = \|y\|_t, \qquad b_\alpha = \frac{y(\alpha)}{R}.$$

Blei's $Q_A^{(t)}(y)$, with the paper's default parameter $\epsilon = 1$, is the formal Walsh series

$$Q_A^{(t)}(y) = R \ \mathrm{Im} \prod_{\alpha \in A} (1 + i b_\alpha r_\alpha).$$

Equivalently, its Walsh coefficients are

$$\widehat{Q_A^{(t)}(y)}(r_F) = \begin{cases} (-1)^k R^{1-(2k+1)} \prod_{\alpha \in F} y(\alpha), & |F| = 2k+1, \\ 0, & |F| \text{ even.} \end{cases}$$

We say that $Q_A^{(t)}(y)$ represents a measure if there is a finite complex Borel measure $\mu \in M(\Omega_A)$ whose Fourier-Stieltjes coefficients agree with these numbers on all Walsh characters.

Theorem

Theorem 1. *Let $2 < t \le \infty$, let A be any set, and let $y \in \ell^t_{\mathbb{R}}(A)$. Then $Q_A^{(t)}(y)$ represents a finite complex measure on Ω_A if and only if $y \in \ell^2(A)$.*

Consequently, if A is infinite and contains a countably infinite subset, there are vectors $y \in \ell^t_{\mathbb{R}}(A)$ for which $Q_A^{(t)}(y)$ does not represent a measure. For instance, on a countable subset take

$$y_n = n^{-1/2} \qquad (n \ge 1),$$

and set $y = 0$ off that subset.

Proof Intuition

The positive direction is the familiar Riesz-product threshold. If $y \in \ell^2$, then the finite complex products $\prod(1 + i b_\alpha r_\alpha)$ have total variation bounded by $\prod(1 + b_\alpha^2)^{1/2} < \infty$, so a weak-* cluster point supplies a measure with the prescribed Walsh coefficients.

The obstruction when $y \notin \ell^2$ is already visible on finite coordinate sets. If a global representing measure existed, integrating the normalized odd part of the finite product against it would be bounded by the total variation of that measure. But the same integral is exactly the squared ℓ^2-mass of the odd Walsh coefficients, divided by the sup norm of the test polynomial. This lower bound grows like $\prod(1 + b_\alpha^2)^{1/2}$, which diverges precisely when $\sum b_\alpha^2 = \infty$.

Proof. The case $y = 0$ is trivial, so assume $y \neq 0$, and write $R = \|y\|_t$ and $b_\alpha = y(\alpha)/R$. Since $t > 2$ or $t = \infty$, we have $|b_\alpha| \le 1$ for all α.

First assume $y \in \ell^2(A)$. For a finite set $B \subset A$, define the absolutely continuous complex measure

$$d\nu_B = \prod_{\alpha\in B} (1 + i b_\alpha r_\alpha)\, d\mathbb{P}_A.$$

Its total variation satisfies

$$\|\nu_B\|_M = \int_{\Omega_A} \prod_{\alpha\in B} |1 + i b_\alpha r_\alpha|\, d\mathbb{P}_A = \prod_{\alpha\in B} (1 + b_\alpha^2)^{1/2} \le \exp\left(\frac{1}{2}\sum_{\alpha\in A} b_\alpha^2\right) < \infty.$$

Thus the net (ν_B), indexed by finite subsets of A, is bounded in $M(\Omega_A) = C(\Omega_A)^*$. By weak-* compactness it has a cluster point ν. For each fixed finite $F \subset A$, the coefficient $\widehat{\nu_B}(r_F)$ is eventually equal to

$$i^{|F|} \prod_{\alpha\in F} b_\alpha.$$

Hence

$$\widehat{\nu}(r_F) = i^{|F|} \prod_{\alpha\in F} b_\alpha.$$

The measure

$$\mu = R\, \mathrm{Im}\, \nu$$

therefore has exactly the Walsh coefficients of $Q_A^{(t)}(y)$. This proves the positive direction.

Conversely, assume that $Q_A^{(t)}(y)$ is represented by a finite measure μ. For a finite $B \subset A$, put

$$P_B = \mathrm{Im} \prod_{\alpha\in B} (1 + i b_\alpha r_\alpha).$$

Write

$$P_B = \sum_{\substack{F\subset B\\ |F| \text{ odd}}} p_F r_F, \qquad p_F = (-1)^{(|F|-1)/2} \prod_{\alpha\in F} b_\alpha.$$

If $P_B \ne 0$, then $f_B = P_B/\|P_B\|_\infty$ is a continuous function with $\|f_B\|_\infty \le 1$. Since the Walsh coefficients of μ on odd $F \subset B$ equal Rp_F, finite-dimensional Fourier duality gives

$$\|\mu\|_M \ge \left|\int_{\Omega_A} f_B\, d\mu\right| = \frac{R}{\|P_B\|_\infty} \sum_{\substack{F\subset B\\ |F| \text{ odd}}} p_F^2.$$

Now

$$\|P_B\|_\infty \le \prod_{\alpha\in B} (1 + b_\alpha^2)^{1/2},$$

and

$$\sum_{\substack{F\subset B\\ |F| \text{ odd}}} p_F^2 = \frac{1}{2}\left(\prod_{\alpha\in B}(1 + b_\alpha^2) - \prod_{\alpha\in B}(1 - b_\alpha^2)\right).$$

Set

$$A_B = \prod_{\alpha\in B}(1 + b_\alpha^2), \qquad C_B = \prod_{\alpha\in B}(1 - b_\alpha^2).$$

The preceding estimates imply

$$\|\mu\|_M \geq \frac{R}{2}\frac{A_B - C_B}{A_B^{1/2}} = \frac{R}{2}A_B^{1/2}\left(1 - \frac{C_B}{A_B}\right).$$

If $y \notin \ell^2(A)$, then

$$\sup_{B\subset A \text{ finite}} \sum_{\alpha\in B} b_\alpha^2 = \infty,$$

and hence

$$\sup_{B\subset A \text{ finite}} A_B = \infty.$$

Moreover,

$$0 \leq \frac{C_B}{A_B} = \prod_{\alpha\in B} \frac{1-b_\alpha^2}{1+b_\alpha^2} \leq \exp\left(-\sum_{\alpha\in B} b_\alpha^2\right),$$

with the convention that the left side is 0 if some $|b_\alpha| = 1$. Along finite sets B with $\sum_{\alpha\in B} b_\alpha^2 \to \infty$, the right side tends to 0, while $A_B^{1/2} \to \infty$. The displayed lower bound for $\|\mu\|_M$ is therefore unbounded, contradicting the finiteness of $\|\mu\|_M$. Thus every representing measure forces $y \in \ell^2(A)$.

Finally, if A contains a countably infinite subset $\{\alpha_n : n \geq 1\}$, the vector $y(\alpha_n) = n^{-1/2}$, $y = 0$ elsewhere, belongs to $\ell^t(A)$ for every $t > 2$ and to $\ell^\infty(A)$, but not to $\ell^2(A)$. The characterization above gives the advertised counterexamples. □

Verification Notes

The proof is analytic. The included script `code/check_lower_bounds.py` evaluates the finite lower bound

$$\frac{R}{2}A_B^{1/2}\left(1 - \frac{C_B}{A_B}\right)$$

for the sample vector $y_n = n^{-1/2}$ and increasing finite sets $B = \{1, \ldots, N\}$. The numerical growth is a sanity check only; it is not used as proof.

Novelty and Literature Check

The run indexes were searched for arXiv:1812.05412, $Q_A^{(t)}$, "represent a measure", "bona fide integrals", and nearby $S(p)$, $L(p)$, Sidon, and $\Lambda(2)$ terminology. No existing solution, attempt, or proof-gap packet covering Question 5.14 was found.

External exact-phrase searches were run for close variants around Blei's terminal questions, including "Does q-Sidon S(q*)", "q-Sidon S(p) Blei", "Does Lambda(2) L(p) Blei", and "Lambda(2)-uniformizable". These searches did not surface a later answer to Question 5.14. Novelty confidence is moderate: the obstruction is elementary once the finite Walsh test is written down, but the bounded search did not reveal a known statement of this exact threshold.

Limitations

This packet answers the measure-existence example in Question 5.14. It does not address the other broad thin-set questions in the source paper, such as $\Lambda(2)$-uniformizability, the $\Lambda(2)$-set problem, the finite union problem for $\Lambda(2)$-sets, or the q-Sidon versus $S(q^*)$ question.

Human Review Recommendation

Reviewers should first check the finite-dimensional duality estimate in the negative direction:

$$\left| \int f_B \, d\mu \right| = \frac{R}{\|P_B\|_\infty} \sum_{|F| \text{ odd}} p_F^2.$$

The second point to check is the positive direction's weak-* compactness argument for the bounded complex Riesz products when $y \in \ell^2$.

References

[1] R. Blei, *From Parseval to Grothendieck and a little beyond, part I*, arXiv:1812.05412, 2018.